Stage 2 Validation Report

Commander v2 — Graduated Deception Architecture

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1. Executive Summary

This report documents the validation of Shadow Tier Stage 2: the Commander v2 Graduated Deception Architecture. The Commander monitors Suricata's behavioral alert stream in real time, tracks unique behavioral conditions per attacker IP, and automatically escalates through three distinct deception states as conditions accumulate — without any operator intervention.

Stage 2 validation confirms the complete three-condition, three-state escalation loop operating end-to-end, evidenced by terminal output screenshots captured during live validation sessions on 5–6 August 2026.

2. Group 1 — Commander Startup

The Commander was launched via:

```
sudo python3 /opt/shadowtier/shadowtier_redirect_v2.py &
```

Commander log confirmed startup sequence — three lines logged immediately on launch:

Evidence 1a	`[2026-08-06 06:17:50] [COMMANDER] Shadow Tier Commander v2 starting`

Evidence 1b	`[2026-08-06 06:17:50] [COMMANDER] Watching: /var/log/suricata/fast.log`

Evidence 1c	`[2026-08-06 06:17:50] [COMMANDER] States: 1=Layer Zero intensify | 2=Challenge gate | 3=Maze redirect`

The Challenge Gate was confirmed running in parallel:

```
[2026-08-04 15:50:53] Shadow Tier Challenge Gate listening on 127.0.0.1:8080
```

3. Group 2 — Three Condition Progression

Attack traffic was generated from the Kali attacker host (192.168.20.2) using hping3 SYN flood and sequential curl POST requests to financial endpoints. The Commander tracked each unique behavioral condition as Suricata fired the corresponding SHADOWTIER rules.

3.1 CONDITION 1/3 — syn_scan

First unique condition triggered by SYN flood traffic. Suricata rule 1000002 fired, Commander extracted attacker IP 192.168.20.2 and logged first condition:

Evidence 2a	[2026-08-06 06:28:58] [COMMANDER] CONDITION 1/3 — 192.168.20.2 — triggered: syn_scan

Supporting Suricata fast.log evidence:

```
[**] [1:1000002:1] SHADOWTIER Corrupted Assumption - SYN scan behavior detected [**]
[Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.20.2:1208 ->
192.168.20.1:80
```

3.2 CONDITION 2/3 — persistence

Second unique condition triggered when Suricata rule 1000001 fired on the attacker returning after silent absorption within the 60-second tracking window:

Evidence 2b	[2026-08-06 06:29:16] [COMMANDER] CONDITION 2/3 — 192.168.20.2 — triggered: persistence

Supporting fast.log evidence:

```
[**] [1:1000001:1] SHADOWTIER Persistence Detected - source returning after silent absorption
[**]
[Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.20.2:1695 ->
192.168.20.1:80
```

3.3 CONDITION 3/3 — login_targeting and beyond

Third unique condition triggered by directed HTTP POST requests to financial endpoints:

```
for i in {1..3}; do curl -s -X POST http://192.168.20.1/login; done
for i in {1..3}; do curl -s -X POST http://192.168.20.1/auth; done
for i in {1..3}; do curl -s -X POST http://192.168.20.1/admin; done
for i in {1..3}; do curl -s -X POST http://192.168.20.1/api; done
```

Evidence 2c	[2026-08-06 06:53:52] [COMMANDER] CONDITION 5/3 — 192.168.20.2 — triggered: api_targeting

Full conditions set confirmed in DEBUG8 output:

Evidence 2d	DEBUG8 — current conditions={'api_targeting', 'persistence', 'syn_scan', 'auth_targeting', 'login_targeting'} state=3

Condition deduplication confirmed by DEBUG9:

```
[COMMANDER] DEBUG9 — condition syn_scan already in set, skipping
```

4. Group 3 — State Transitions

Each state transition was triggered automatically by the Commander as unique condition counts accumulated. No operator action was taken between detection and state activation.

4.1 STATE 1 — Layer Zero Intensification

Evidence 3a	[2026-08-06 06:28:58] [COMMANDER] STATE 1 — 192.168.20.2 — activating Layer Zero intensification

iptables rules injected automatically:

```
iptables -t nat -A PREROUTING -s 192.168.20.2 -p tcp --dport 22 -j DNAT --to-destination
192.168.20.1:9999
iptables -t nat -A PREROUTING -s 192.168.20.2 -p tcp --dport 3306 -j DNAT --to-destination
192.168.20.1:9998
```

4.2 STATE 2 — Challenge Gate Routing

Evidence 3b	[2026-08-06 06:29:16] [COMMANDER] STATE 2 — 192.168.20.2 — routing to challenge gate

iptables rule injected automatically:

```
iptables -t nat -A PREROUTING -s 192.168.20.2 -p tcp --dport 80 -j DNAT --to-destination
192.168.20.1:8080
```

4.3 STATE 3 — Maze Redirect

Evidence 3c	[SHADOW TIER] Attacker detected: 192.168.20.2 — redirecting to maze

Evidence 3d	[SHADOW TIER] 192.168.20.2 now inside the maze

iptables rule injected automatically:

```
iptables -t nat -A PREROUTING -s 192.168.20.2 -p tcp --dport 80 -j DNAT --to-destination
192.168.30.2:80
```

5. Validation Summary

Component	Result	Evidence Reference
Commander v2 startup	**PASS**	Evidence 1a/1b/1c — startup sequence confirmed in commander.log
fast.log watching confirmed	**PASS**	Evidence 1b — Watching path logged on startup
CONDITION 1/3 — syn_scan	**PASS**	Evidence 2a — logged at 06:28:58, SID 1000002
CONDITION 2/3 — persistence	**PASS**	Evidence 2b — logged at 06:29:16, SID 1000001
CONDITION 3+ — HTTP endpoints	**PASS**	Evidence 2c/2d — login, auth, admin, api all triggered
STATE 1 activation	**PASS**	Evidence 3a — Layer Zero intensification logged
STATE 2 activation	**PASS**	Evidence 3b — Challenge gate routing logged
STATE 3 — Maze redirect	**PASS**	Evidence 3c/3d — attacker inside the maze confirmed
Condition deduplication	**PASS**	DEBUG9 — repeat conditions correctly skipped
No operator intervention	**PASS**	Full escalation automatic from detection to maze
Challenge Gate startup	**PASS**	challenge.log — listening on 8080 confirmed

6. Conclusion

Stage 2 of Shadow Tier is validated. The Commander v2 Graduated Deception Architecture demonstrated a complete, automated behavioral escalation pipeline across three states, driven by six unique behavioral conditions triggered by real attack traffic from a Kali Linux attacker host.

— All three state transitions activated automatically without operator intervention.

— Five unique behavioral conditions tracked and logged across the validation session.

— Condition deduplication correctly prevented repeat triggers from inflating condition counts.

— Source-specific iptables DNAT rules injected per state, targeting only the confirmed attacker IP.

— Attacker confirmed inside the deception maze via redirect engine log output.

Combined with Stage 1 XDP validation, Shadow Tier demonstrates a complete four-layer active deception pipeline from volumetric flood absorption to full deception maze engagement, all automated, all without operator intervention.